

The Utility Problem

By design, the electricity distribution grids are built as a 3-phase. If some loads are connected to only one phase, or 3-phase loads have more load on one phase, this results in a change in the balance of power flow on each phase. Because of this, one phase in the distribution grid may be more loaded than the other two phases, or one phase may be significantly less loaded than the other two phases.

Such a 3-phase power imbalance then causes several undesirable situations. For example, due to the imbalance, the maximum carrying capacity of the line can be affected. This happens even when the other phases have sufficient reserve capacity. Another undesirable consequence of imbalance can be an increase in losses due to the flow of currents through the neutral conductor, which affects the economics and billing of the grid operation.

In the past (without the use of data analytics), this situation would have been solved by upgrading the line with a higher diameter line, which would have caused high investment costs. By utilizing data, it is possible to identify which phase is more or less loaded, and it is possible to switch some loads from this phase to another one, thus optimizing the grid, where such a solution presents much lower investment costs compared to upgrades of the conductors.

The Data Solutions and Methods

For conducting such an analysis, it is necessary to have time-series data of the current for each phase. With the Awesense Platform, it is very easy to analyze the phase load either in the entire grid or only in selected parts of the grid. The unique integration of geospatial data (GIS) with time series data (sensor data – current measurement) in the Awesense Platform enables this type of analysis. Pre-built tracing functions speed up the creation of analyses for selected parts of the grid. Any phase balance result can be displayed for a given part of the grid (e.g. sum of imbalances by phases, average, median, maximum, etc.). These values can then be plotted on a map or table for a more comprehensive evaluation.

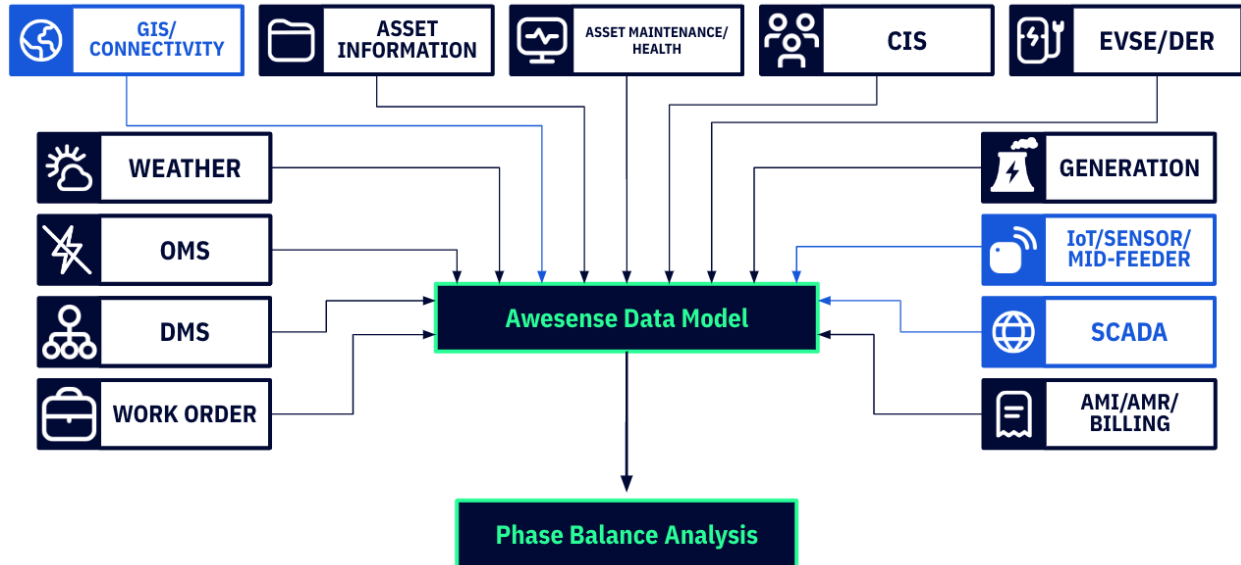
In addition to developing custom phase balance analyses using the Awesense EDM interface, the Awesense TGI software provides phase balance reporting. This includes information about a single location on the grid instantly and without the need to write new code. The grid wide overview of the power factor is available as well for better understanding of the overall power factor situation.

Required Data Sets

The required data for this use case are visualized in the following diagram. Data sources highlighted in [royal blue](#) are required for this use case.

Phase Balance Analysis

[UC03-02] - Awesense Energy Transition Platform Use Case



Available Use Case Implementations:

Users of the Awesense Energy Transition Platform can access this use case by the following means:

1. TGI Phase Balance Analysis - Please speak to Awesense about access to the sandbox environment or if you would like to see a demo of the solution.